MELVIN TABITHA OPONDO

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PROFESSIONAL SUMMARY

I am a full-stack Software Developer with hands on experience building responsive web applications using React.js, Django, and Flask. I am proficient in designing RESTful APIs, managing relational and non relational databases, and deploying scalable systems. I work across the full development lifecycle, from architecture and backend logic to frontend interfaces and API integration. I am committed to clean code, continuous learning, and contributing to impactful real world projects.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, HTML, CSS

Frontend: React.js, React Native, TailwindCSS, Bootstrap, Vite, React Hook Form, React Router

Backend & Frameworks: Django, Django REST Framework, Flask, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis

Tools & DevOps: Git, GitHub, Docker, Kubernetes (learning), Postman, Ngrok, Vercel, Render, Neon, Oracle Cloud, Netlify, Firebase

APIs & Integrations: RESTful API development, Safaricom Daraja (M-Pesa), Google Maps API, OpenWeather, JWT Authentication

Testing: Pytest, Django Test Framework, React Testing Library, Jest

Other: Technical SEO, Canva, Adobe Photoshop

WORK EXPERIENCE

Computer Science Academy Africa (CSA)
Web Developer, Website Maintenance & Technical SEO | csaafrica.org

- I designed, developed, and deployed the live website for Computer Science Academy Africa, configuring its production hosting on the organisation's accounts and implementing the technical SEO fundamentals that keep the organisation discoverable to the students and partners it exists to reach.
- I take responsibility for a real external client whose users depend on the site being up and findable, not a weekend project. I handle deployment, monitoring, and ongoing content and performance updates.
- I apply technical SEO practices (clean site structure, crawlable markup, descriptive metadata, performance oriented page delivery) so the organisation reaches its audience through organic search rather than paid acquisition alone.

ACSIS, African Caribbean Sustainability & Investment Summit
Web and Graphic Designer

- I designed graphics and visual materials that supported the organisation's communications, branding, and visibility.
- I worked collaboratively with a team of designers, coordinating on shared design work so that each piece aligned with the organisation's branding requirements.
- I contributed to keeping the organisation's materials visually consistent and to improving the presentation of its communications.

PROJECTS

Anga: Weather Platform for Kenya
Live | *Code*

- As sole developer, I engineered a weather platform for 99 Kenyan counties and towns around a strict upstream limit of 1,000 API requests per month, placing a Django REST API between the React frontend and the Weather AI API so that user requests do not become upstream calls.
- I implemented two-tier Redis caching (30 minute fresh window, 6 hour stale fallback) and request coalescing with atomic Redis locks, so concurrent cache misses trigger a single upstream call. Verified on a running server: 30 simultaneous requests produced exactly 1 upstream request.

- I built quota protection for the monthly limit: a circuit breaker that serves cached data after a 429, a reserve that halts upstream calls when fewer than 25 requests remain, local quota tracking synced daily against the provider's usage endpoint, and per-client throttling.
- I served location search from a PostgreSQL gazetteer (47 counties, 52 towns with aliases), kept the API key server side behind a CORS allow-list and input validation, and wrote 222 automated tests (98 backend, 124 frontend), including a 25-thread concurrency test on the coalescing lock.
- Stack: Django, Django REST Framework, Redis, PostgreSQL (Neon), React, Vite; deployed on Vercel and Render.

StudyBuddy: AI-Powered Study Assistant

Live | *Code*

- As sole developer, I built an AI study assistant that turns uploaded notes (PDF, Word, plain text, or lecture recordings transcribed with Gemini) into summaries, quizzes, and flashcards grounded strictly in the student's own material.
- I implemented a post-generation verification layer: keywords must appear in the source, definitions and formulae are dropped below 50% and 60% word overlap, and every quiz question must quote an evidence sentence from the notes. Gemini runs at low temperature with structured JSON output.
- I designed the AI layer around a free quota of roughly 20 requests a day: errors classified by type with backoff on transient failures only, automatic fallback across Gemini models, and SHA-256 keyed result caching so identical notes are summarised once.
- I built custom JWT authentication with refresh-token rotation and blacklisting, Argon2 password hashing, constant-time login to prevent account enumeration, server-side Google token verification, per-scope rate limits, and file-signature upload validation, across a REST API of about 31 endpoints.
- Stack: Django, Django REST Framework, PostgreSQL (Neon), Gemini API, React 19, TanStack Start, Tailwind CSS, Cloudinary, Brevo; deployed on Vercel and Render.

CallShield: AI-Powered Scam Call Protection System

- I built an AI-driven call security system that detects and flags potential scam calls in real time, addressing a problem that affects millions of mobile phone users across Africa every day. The product's job is to give a user a signal before they engage with a scam, not after they have lost money.
- I used pattern recognition and data analysis to classify incoming calls, so the model's output can be acted on in the moment rather than after a long investigation.
- I developed the Django backend with RESTful APIs to handle call data processing and machine learning model integration.
- I designed and managed the PostgreSQL database for secure storage of call records and user data.
- I built the frontend using React Native, delivering a responsive and user-friendly mobile experience.
- I served as Project Manager, leading planning, coordinating team tasks, managing timelines, and overseeing testing and deployment readiness. The combination of technical ownership and team leadership reflects the realities of small engineering teams, where the person who ships the code is often also the person who coordinates it.

SmartHunt: Student House Hunting Platform

- I built a student-focused housing platform that connects tenants with verified rental listings and processes B2B payments through the Safaricom Daraja (M-Pesa) API. The project solves a specific real world problem: students arriving in a new city need somewhere safe to live, and the platform integrates with the mobile money infrastructure almost every Kenyan already uses rather than asking them to adopt yet another payment method.
- I developed the backend using Django and Django REST Framework, with FastAPI supporting high-performance services, and designed a MySQL database covering property listings, users, and transactions.
- I integrated the Google Maps API for location-based property discovery, so students can filter listings by proximity to campus rather than scrolling through unrelated options.
- I developed the React frontend, delivering a responsive and intuitive user interface across devices.

UjamaaFlow: Resource Donation Web App

Live | *Code*

- I built a resource donation platform in Flask that connects donors with recipients through a three-role model (donor, recipient, administrator), with role-based access and a separate dashboard for each role.
- I modelled donations, media uploads, requests, matches, and community reports with SQLAlchemy, supporting photo listings across six categories, category and location filtering, admin matching of donations to requests, and a moderation queue fed by community reports.
- Stack: Flask, Flask-SQLAlchemy, Flask-Login, Jinja, SQLite, Bootstrap; deployed on Render.

Master Photography Kenya: Studio Website

Live | *Code*

- I built a ten-page studio website in HTML, CSS, and vanilla JavaScript that makes no third-party requests, with self-hosted fonts, a filterable gallery with lightbox, and structured data (ProfessionalService, FAQ, breadcrumbs) for search.
- I wrote a Python static site generator that renders every page from shared partials, and an audit script that fails the build on broken internal links, missing alt text or image dimensions, invalid JSON-LD, malformed HTML, and unverified business claims.
- I implemented a WhatsApp enquiry flow with custom validation of Kenyan phone numbers and a honeypot field against bots, composing each enquiry as a pre-filled message to the studio.
- Stack: HTML, CSS, JavaScript, Python (build and audit tooling), JSON-LD; deployed on Vercel.

Developer Portfolio

Live

- I built my portfolio as a static site with no framework or build step, working identically over HTTP or from the local file system.
- I implemented an interactive terminal (command parsing, history, tab completion) as a progressive enhancement, light and dark themes applied before first paint, and accessible defaults including a skip link, ARIA on interactive controls, and reduced-motion support.
- I added SEO foundations: canonical URLs, a sitemap, structured data, and Open Graph tags.
- Stack: HTML, CSS, JavaScript; deployed on Vercel.

Weatherfusion: Real-Time Weather Application

Live | *Code*

- I built a Django weather application that fetches current conditions and forecasts from the OpenWeatherMap API for any city, displaying temperature, humidity, and pressure.
- I integrated a conversational assistant on Groq’s Llama 3.1 model, injecting the live OpenWeatherMap readings for the requested city into the prompt so answers are based on current data rather than the model’s own knowledge.
- Stack: Django, OpenWeatherMap API, Groq (Llama 3.1), HTML, CSS; deployed on Render.

EDUCATION & TRAINING

Meru University of Science and Technology

Bachelor of Science in Computer Science

Computer Science Academy (CSA)

Software Engineering

Power Learn Project (PLP)

Software Engineering

Africa Agility Foundation

Frontend Web Development

Workmates Technology

UI/UX Design

ALX, AI Career Essentials Program

AI tools integration and ethical AI applications in industry workflows

Kolanya Girls’ School

KCSE | 2019 to 2022

LEADERSHIP & VOLUNTEER EXPERIENCE

Community Outreach Specialist. WISSA, Women in Sustainable STEM Africa

- I supported outreach programmes aimed at empowering women and girls in STEM fields across Africa.

Web & Graphic Designer (Volunteer). ACSIS, African Caribbean Sustainability & Investment Summit

- I designed digital graphics and promotional materials using Adobe Photoshop and Canva.
- I worked with other designers to keep event materials consistent with the organisation's branding.

Mentor & STEM Advocate. Kolanya Girls' Secondary School

- I mentored students in STEM career pathways, leadership development, and facilitated coding workshops.

Social Media Manager. Blockchain Community, Meru University Science Innovators Club

- I managed and grew the community's social media presence, creating technical and educational content on blockchain and emerging technologies.
- I supported community engagement, event promotion, and new member onboarding.

SOFT SKILLS

Leadership & Responsibility • Problem Solving & Analytical Thinking • Team Collaboration • Adaptability & Continuous Learning • Effective Communication

LANGUAGES

English, Fluent | Kiswahili, Fluent | French, Intermediate

REFEREES

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